

Kinetics and drug stability

Chapter outline

- objective
- Introduction
- Rate and order of reactions
- Chemical and physical degradation
- Factors affecting stability of drugs
- Stability study
- Prediction of shelf life

objectives

- **At the end of this chapter , you will be able to**
 - ✓ explain chemical kinetics, stability, shelf-life, half-life of pharmaceuticals
 - ✓ calculate concentration, rate constant , half life and shelf-life of drugs ,and determine order of a reaction
 - ✓ discuss physical and chemical degradation of drugs and the factors affecting the stability of drugs
 - ✓ Discuss the types stability study
 - ✓ Predict shelf-life of drugs

Introduction

- **Kinetics** is study of the rate at which processes or changes occur
 - The changes may be
 - **chemical** such as decomposition of a drug, radiochemical decay
 - **physical** such as transfer across a boundary, such as the intestinal lining or skin

Introduction...

Importance of studying kinetics

- **To determine half life**
 - the time necessary for a drug to decay by **50%**
- **To determine shelf life ($t_{0.9}$)**
 - the time necessary for the drug to decay to **90%** of its original concentration.
 - **Expiration date** is the date placed on the container label of a drug product designating the time prior to which a batch of the product is expected to remain within the approved shelf-life.
- **Selection of proper storage condition**
 - Temperature and humidity
 - advising patient on storage conditions

Introduction...

- Selection of proper container for dispensing
 - Glass vs. plastic
 - Clear vs. amber
- Anticipation of interactions when mixing and in dosage forms (incompatibilities)
- Dissolution determinations
- To study ADME Processes in pharmacokinetics
 - absorption, distribution, metabolism/ biotransformation & excretion

Introduction...

- Stability is the capacity of a drug product to remain within specifications established to ensure its **identity, strength, quality, and purity**
- **Stability study** is defined as the study of the extent to which the **properties of a drug substance or drug product** remain within specified limits at certain temperature and humidity
 - The **properties** may be **physical, chemical, microbiological,** or performance properties such as **disintegration** and **dissolution.**

Introduction.....

Types of stability	Conditions to be maintained throughout the shelf life of the drug product
Chemical	Each active ingredient retains its chemical integrity and labeled potency, within specified limits.
Physical	The original physical properties, for example, appearance, palatability, dissolution ,suspendability , etc are retained.
Microbiologic al	Sterility or resistance to microbial growth is retained according to the specified requirements. Antimicrobial agents that exist retain effectiveness within the specified limits.
Therapeutic	The therapeutic effect remains unchanged.
Toxicological	No significant increase in toxicity occurs.

Introduction...

Importance of stability

- Extensive chemical degradation result in a substantial loss of quantity of therapeutic agent in the dosage form
- Degradation products may result in adverse events or be unsafe
- Instability may cause
 - ✓ Undesired change in performance, i.e. dissolution/bioavailability
 - ✓ Substantial changes in physical appearance of the dosage form causing product failures

Rate and order of reactions

Rate of reaction

- Rate: is the velocity or speed of a reaction with which a reactant(s) undergo a change.
- Can be measured by measuring the change in the conc. of a reactant or product in a particular period of time
- Given by; rate = $\pm dc/dt$

Rate and order of reactions...

Order of reaction

- The **order of a reaction** refers to the way in which the concentration of the reactant (s) influence the rate of a chemical rxn
 - **Zero order rxn:** the rate of rxn is independent of the concentration of the reactants and constant with respect to time

Example.

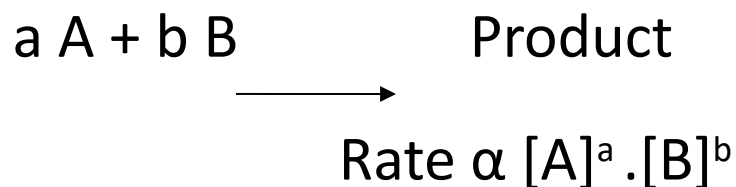
- The concentration of steroid remaining in a transdermal patch is as follows:
- | | | | | | | |
|-------------|----|------|------|-----|-----|-----|
| Time (h) | 0 | 30 | 60 | 90 | 120 | 150 |
| Amount (mg) | 20 | 16.4 | 12.8 | 9.2 | 5.6 | 2 |

Rate and order of reactions...

- **First-order rxn:** the rate of rxn is directly proportional to the concentration reactant remaining with respect to time
- Most commonly, **zero-order and first-order** reactions are encountered in pharmacy

Rate and order of rxn...

○According to law of mass action, the rate of a reaction is proportional to the product of the molar concentration of reactants each raised to power equal to the number of molecules undergoing reaction.



$$\text{Rate} = K [A]^a \cdot [B]^b \quad (1)$$

Rate and order of rxn...

Order of reaction = sum of exponents

Order of A = a and B = b

Then overall order = $a + b$

K is the rate constant or specific rate constant

Rate and order of rxn...

Ex: The reaction of acetic anhydride with ethyl alcohol to form ethyl acetate and water



$$\text{Rate} = K [(\text{CH}_3 \text{ CO})_2 \text{ O}] \cdot [\text{C}_2\text{H}_5\text{OH}]^2$$

Order for $(\text{CH}_3 \text{ CO})_2 \text{ O}$ is 1st order

Order for $[\text{C}_2\text{H}_5\text{OH}]^2$ is 2nd order

Overall order of reaction is 3rd Order

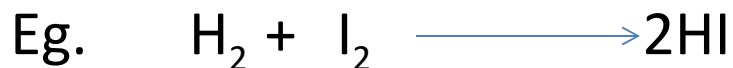
Rate and order of rxn...

Molecularity

- Refers to the number of molecules, ions or atoms in elementary reaction to give the product
- **Unimolecular**: only one type of molecule undergoes a change to yield the product



- **Bimolecular**: two molecule combine to give the product



Rate and order of rxn...

Specific rate constant (k)

- The constant, k , appearing in rate law associated with a single-step rxn
- Any change in the condition of the rxn will lead to rate law having a different value for k eg temperature, solvent,
 - Experimentally, a change of k corresponds simply to a change in the slope of the line given by the rate equation

Rates and orders of reactions

zero order reaction

- It is a reaction where reaction rate is **not** dependent on the concentration of the reacting species
- Rate is constant



$$\text{Rate} = - dC/dt = K [A]^0$$

$$- dc/dt = k$$

$$dc = - k dt$$

where dC is the change in concentration wrt change in time t

‘-’ sign indicates that the concentration is decreasing

Rates and orders of reactions...

- The rate equation may be integrated b/n the initial concentration (C_o) and concentration C_t after time t .

$$\int_{c=c_o}^{c=c_t} dc = \int_{t=0}^{t=t} -k dt$$

$$[c]_{c_o}^{c_t} = [kt]_0^t$$

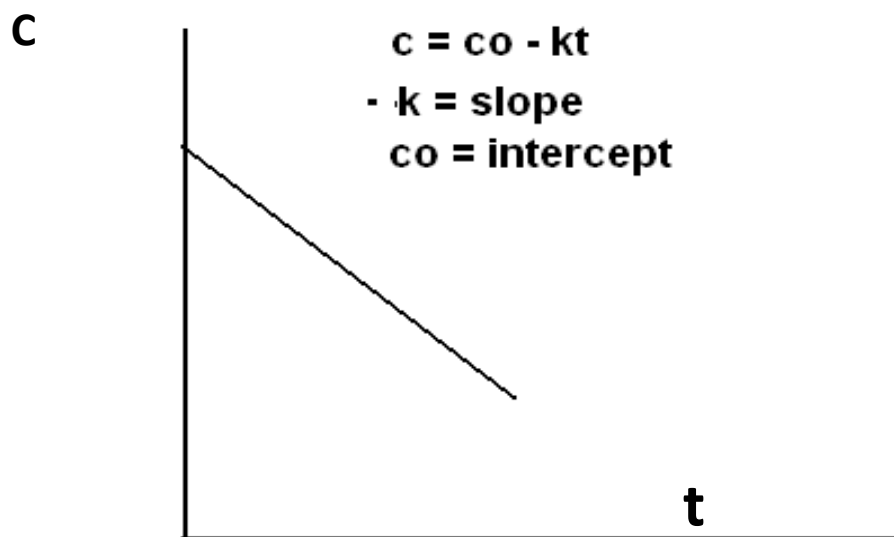
$$c_t - c_o = -kt$$

$$c - c_o = -kt$$

$$C = C_o - kt \quad (2)$$

Rates and orders of reactions

- When this linear equation is plotted with conc on the vertical axis and time on the horizontal axis, the slope of the line give $-K$



- Unit of K for zero order rxn is $\text{moles Litre}^{-1} \text{ second}^{-1}$

Rates and orders of reactions...

Half life of zero order reaction

- The half life is the time required for 50% of the material to disappear; it is the time at which C_0 is decreased to $1/2C_0$

$$\text{Let } C = C_0 / 2 \text{ and } t_{1/2} = t$$

substituting this in to eq. 2, yields

$$C = C_0 - k t$$

$$t_{1/2} = C_0 / 2K \quad (3)$$

Shelf life of a zero order reaction

- The shelf life is the time required for 10% of the material to disappear; it is the time at which C_0 has decreased to 90% of its original concentration

Let $c = 0.9C_0$ and $t = t_{0.9}$

substitute in eq. 2;

$$c = c_0 - k t$$

$$t_{90\%} = t_{0.9} = 0.1 c_0 / k \quad (4)$$

Rates and orders of reactions...

- Q. 1. Drug X degrades by a zero-order process with a rate constant of $0.05 \text{ mg/ml}\cdot\text{year}$ at room temperature. If a 1 mg/ml solution is prepared and stored at room temperature:
1. What concentration will remain after 18 months? Ans 0.925
 2. What is the half-life and shelf-life of the drug? Ans 10 and 2 years
- Q. 2. The conc of a drug preparation exposed at 40°C decreased from 1.420 mg/ml to 1.400 mg/ml in 3 months. Determine the rxn rate assuming that the rxn follows zero order rxn. Ans $0.00667 \text{ mg/ml} \cdot \text{Mon}$

Rates and orders of reactions...

First-order reaction

- The rate of the rxn is directly proportional to the first power of a concentration of single reactant.
- The reaction can be written as



- The rate equation is given as

$$-\frac{dc}{dt} = kc$$

(5)

Where C is the concentration of the reactant A remaining undecomposed at time t and K is the first-order rate constant

Rates and orders of reactions...

- Integrating this equation b/n the limits of concentration C_0 at time $t=0$ and concentration C at $t=t$, we get

$$\int_{C_0}^C \frac{dc}{c} = -k \int_0^t dt$$

$$\ln c - \ln C_0 = -k(t - 0)$$

$$\ln c = \ln C_0 - kt$$

(6)

- Converting to a logarithmic form, we get

$$\log c = \log C_0 - kt/2.303$$

(6)

or

$$k = \frac{2.303}{t} \log \frac{C_0}{c}$$

(7)

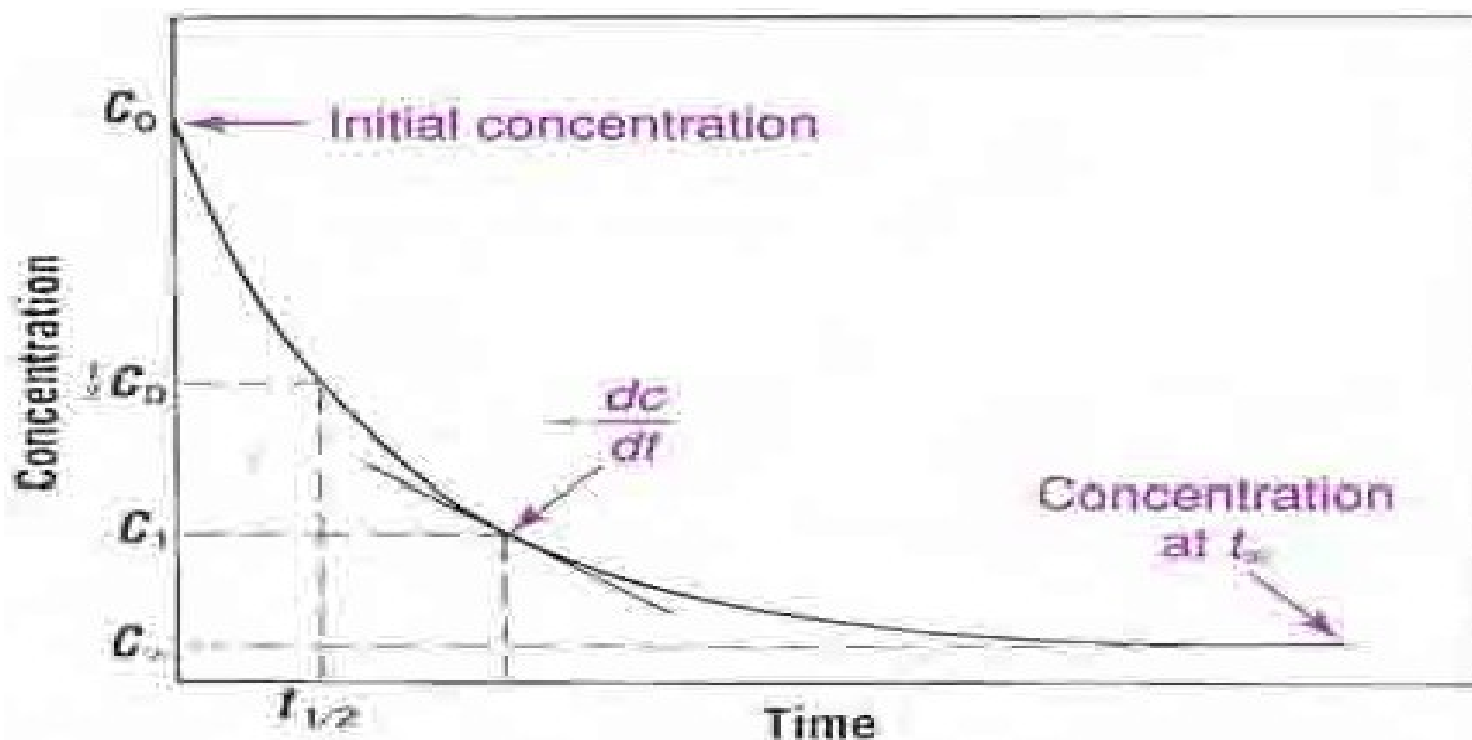
Rates and orders of reactions...

- In exponential form, the equation becomes

$$c = c_0 e^{-kt} \quad (8)$$

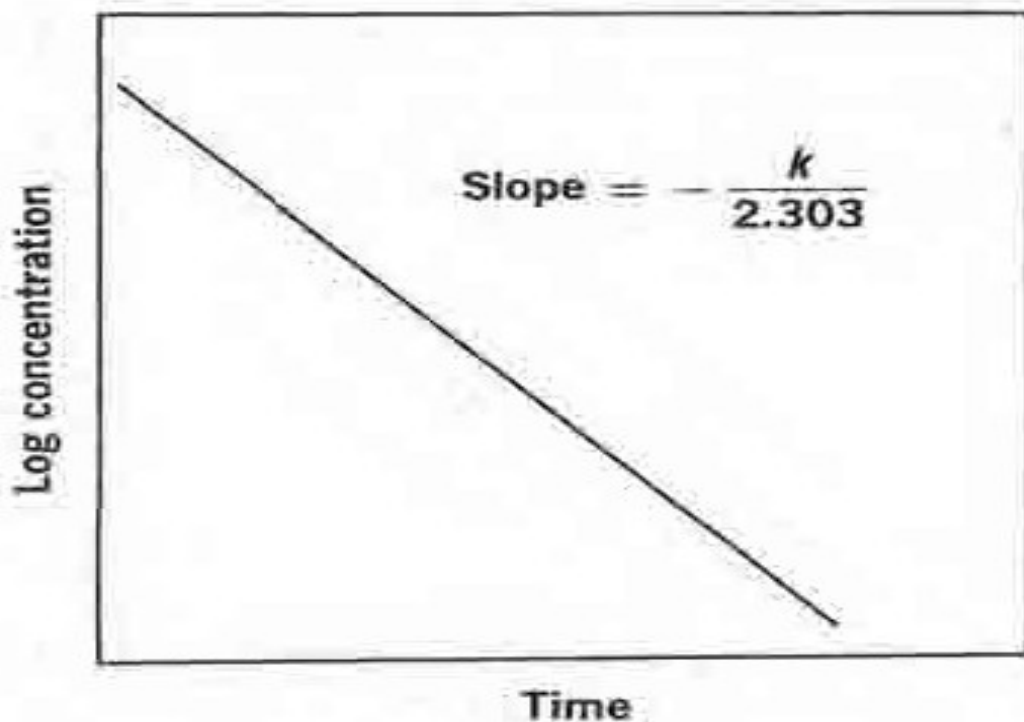
$$\text{or } c = c_0 10^{-kt/2.303} \quad (9)$$

- The conc decrease exponentially with time



Rates and orders of reactions...

- If $\log c$ is plotted against time t , a straight line is obtained with slope equal to $-k/2.303$.
- The rate constant k can be obtained from the slope of the line and it has unit of Sec^{-1}



Rates and orders of reactions...

First-order reaction eg..

- Consider the following. A cardiac stimulant is administered by i.v. injection. Blood samples have the following radioactivity counts per second (cps)(C)

• $t \text{ (min)}$	0	30	60	90	120	150
• C	59.7	24.3	9.87	4.01	1.63	0.67
• $\ln (C)$	4.09	3.19	2.29	1.39	0.49	0.41

Calculate the first order rate constant ,K ?

Rates and orders of reactions...

Half life of first order reaction

$$C = 1/2 C_0$$

substituting in the equation

$$t = \frac{2.303}{k} \log C_0/C$$

k

$$t_{1/2} = 0.693/k \quad (10)$$

Rates and orders of reactions...

Shelf life of first order reaction

$$C=0.9C_0$$

substituting in equation

$$t = \frac{2.303}{k} \log C_0/C$$

k

$$t_{90\%} = 0.105/k \quad (11)$$

- the half life and shelf life of first order rxn is a constant independent of the concentration

Rates and orders of reactions...

Q. 1. A solution of drug contained 680 units/ml when prepared. It was analysed after a period of 60 days and was found to contain 400 units/ml. assuming the decomposition is first order, at what time the drug have decomposed to one half of its original concentration? Ans 78 days and 7 hrs

Q.2. Based on the Q 1 what the time taken for 5% of a drug to decompose?

Rates and orders of reactions...

Q.3. The initial concentration of active principle in an aqueous preparation was $5.0 \times 10^{-3} \text{ gcm}^{-3}$. After 20 months the concentration was shown by analysis to be $4.2 \times 10^{-3} \text{ gcm}^{-3}$. The drug is known to be ineffective after it has decomposed to 70% of its original concentration. Assuming that decomposition follows first order kinetics, calculate the expiry date of the drug preparation.

Q. 4. The rate of decomposition of a 0.056 M glucose at 140 oC in acidic solution was found to be as follows based on this data fined K, half life and shelf life.

Time (in hrs)	0.0	0.5	2.0	4.0	6.0	8.0	12.0
Glucose remaining ($\times 10^{-2} \text{ mol.liter}^{-1}$)	5.65	5.52	5.31	5.02	4.80	4.52	4.09

Rates and orders of reactions...

Second order reaction

- There are two forms of second order reaction

Type I: $2A \rightarrow \text{Product}$

e.g., decomposition of hydrogen peroxide



Type II: $A + B \rightarrow \text{Product}$

e.g., saponification of ethyl acetate



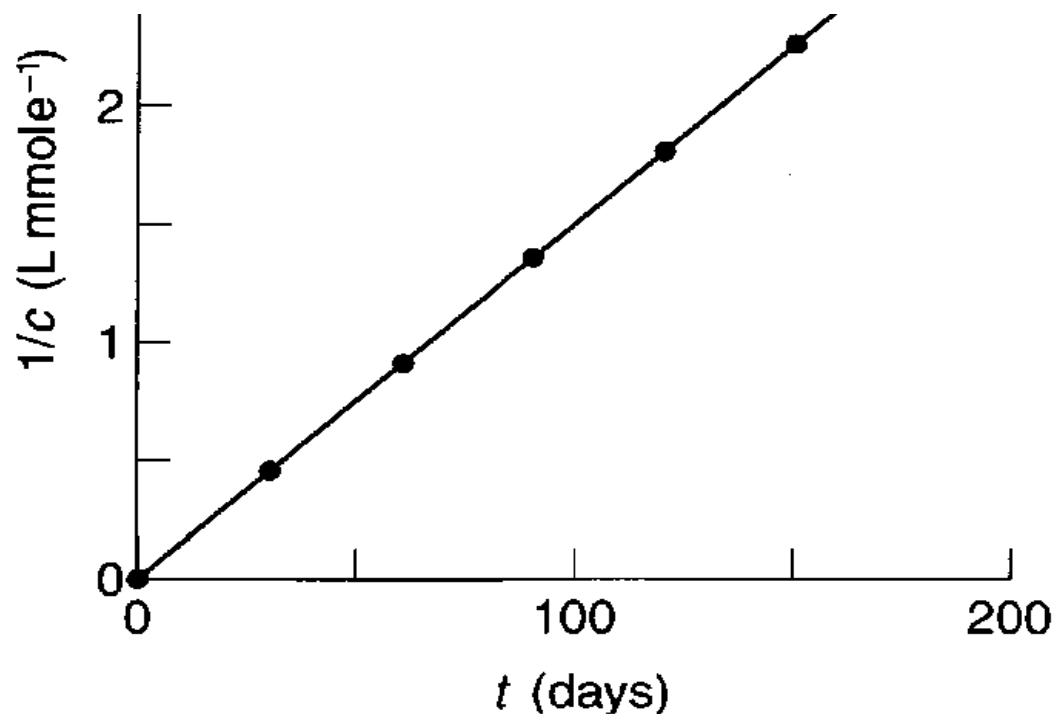
Rates and orders of reactions...

Second order reaction

Plot of $1/c$ vs. t or $x/a(a-x)$ against t

$$\frac{-dC}{dt} = KC^2$$

$$\int_{C_0}^{C_t} \frac{dC}{C^2} = -K \int_{t=0}^{t=t} dt$$



Rates and orders of reactions...

Second-order reaction

For **Type II: $A + B \rightarrow \text{Product}$**

- If the speed of the reaction depends on the concentrations of A and B with each term raised to the first power, the rate of decomposition of A is equal to the rate of decomposition of B, and both are proportional to the product of their concentrations

Rates and orders of reactions...

$$-\frac{d[A]}{dt} = \frac{d[B]}{dt} = k[A][B] \quad (14)$$

- If a and b are the initial concentrations of A and B, respectively, and x is the concentration of each species reacting in $\frac{dx}{dt} = k(a-x)(b-x)$ rate is written as

$$(15)$$

- where $a-x$ and $b-x$ are the concentrations of A and B, respectively, remaining at time t .

Rates and orders of reactions...

- If both A and B present in the same concentration so that $a=b$ the equation is written as

$$\frac{dx}{dt} = k(a - x)^2 \quad (16)$$

- On integrating b/n the limits $x=0$ at $t=0$ and $x=x$ and at $t=t$

$$\int_0^x \frac{dx}{(a - x)^2} = k \int_0^t dt$$

$$\left(\frac{1}{a - x} \right) - \left(\frac{1}{a - 0} \right) = kt$$

$$\frac{x}{a(a - x)} = kt \quad (15) \quad \text{or}$$

$$k = \frac{1}{at} \left(\frac{x}{a - x} \right) \quad (17)$$

Rates and orders of reactions...

- Plot of $x/a(a-x)$ against t yields straight line and the slope of the line is K
- If A and B are not present in equal concentration, integration of eq. 15 yields

$$\frac{2.303}{a-b} \log \frac{b(a-x)}{a(b-x)} = kt \quad (18)$$

or

$$k = \frac{2.303}{t(a-b)} \log \frac{b(a-x)}{a(b-x)} \quad (19)$$

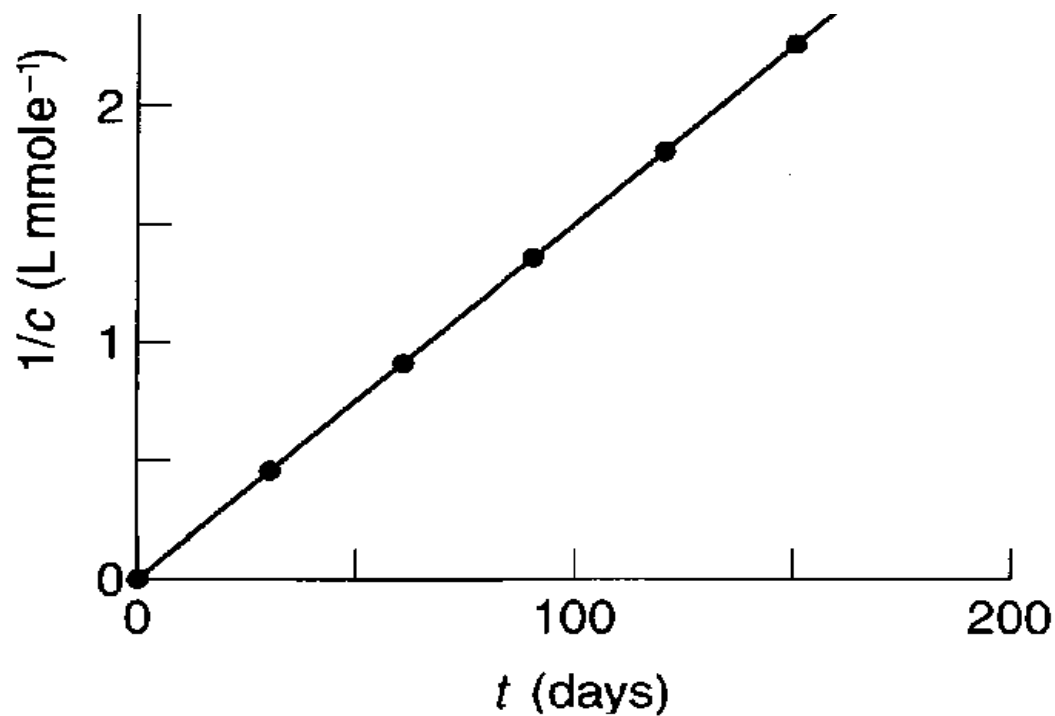
Rates and orders of reactions...

Second order reaction

Plot of $1/c$ vs. t or $x/a(a-x)$ against t

$$\frac{-dC}{dt} = KC^2$$

$$\int_{C_0}^{C_t} \frac{dC}{C^2} = -K \int_{t=0}^{t=t} dt$$



Rates and orders of reactions...

- plot of $\log[b(a-x)/a(b-x)]$ against t yields straight line with a slope of $(a-b)k/2.303$
- If one species is involving in the reaction a plot of $1/c$ vs. t gives a straight line with a slope k .
- The constant has a unit litre/mole.sec

Rates and orders of reactions...

Half life of second order reaction

➤ If $a=b$

$$t_{1/2} = \frac{1}{ak} \quad (20)$$

➤ If a is not equal b and when $x=0.5a$

$$\frac{1}{k(a-b)} \ln \frac{0.5ab}{a(b-0.5a)} \quad (21)$$

➤ If a is not equal b and when $x=0.5b$

$$\frac{1}{k(a-b)} \ln \frac{b(a-0.5b)}{0.5ab} \quad (22)$$

Rates and orders of reactions...

Shelf life of second order reaction

➤ If $a=b$

$$\frac{0.11}{c_0 k} \quad (23)$$

➤ If a is not equal b and when $x=0.1a$

$$\frac{1}{k(a-b)} \ln \frac{0.9ab}{a(b-0.1a)} \quad (24)$$

➤ If a is not equal b and when $x=0.1b$

$$\frac{1}{k(a-b)} \ln \frac{b(a-0.1b)}{0.9ab} \quad (25)$$

Rates and orders of reactions...

Q. 1. In a saponification rxn of ethyl acetate at 25 °C, the concentration of sodium hydroxide remaining after 1 hr was 0.00684 M if the initial conc of the ester as well as the base was 0.01 M, calculate the second order rxn rate constant and half life and shelf life of the rxn. Ans $k=0.76998$ liter/mole.min, $t_{1/2}= 130$ min and $t_{90\%}$ 14 min

Rate expression for the zero, first and second order reaction

	Zero-order	First-order	Second-order	
			$a = b = c_0$	$a \neq b$
Differential rate expression	$-\frac{dc}{dt} = k$	$-\frac{dc}{dt} = kc$	$-\frac{dc}{dt} = kc^2$	$-\frac{dc}{dt} = kc_a c_b$
Integrated rate expression	$k = \frac{c_0 - c}{t}$	$k = \frac{1}{t} \ln \frac{c_0}{c}$	$\frac{1}{c} - \frac{1}{c_0} = kt$	$k = \frac{1}{t(a-b)} \ln \frac{b(a-x)}{a(b-x)}$
$t_{1/2}$	$\frac{c_0}{2k}$	$\frac{0.693}{k}$	$\frac{1}{c_0 k}$	(i) When $x = 0.5a$ $\frac{1}{k(a-b)} \ln \frac{0.5ab}{a(b-0.5a)}$ (ii) When $x = 0.5b$ $\frac{1}{k(a-b)} \ln \frac{b(a-0.5b)}{0.5ab}$
$t_{90\%}$	$\frac{c_0}{10k}$	$\frac{0.105}{k}$	$\frac{0.11}{c_0 k}$	(i) When $x = 0.1a$ $\frac{1}{k(a-b)} \ln \frac{0.9ab}{a(b-0.1a)}$ (ii) When $x = 0.1b$ $\frac{1}{k(a-b)} \ln \frac{b(a-0.1b)}{0.9ab}$

Rates and orders of reactions...

Pseudo order reaction

- **Apparent or pseudo order** describes a situation where one of the reactants is present in large excess or does not effect the overall reaction and can be held constant

Pseudo zero order reaction

- In first-order rxn, if the concentration of reactant is remains constant, then the rate of change of concentration follows zero-order.

Rates and orders of reactions...

- A compound decomposing in solution exhibits a first order rxn but if more of it is present as insoluble excess, then an equilibrium is maintained b/n the compound in solution and that in the solid form
- The concentration of the compound in solution, thus remains essentially constant till any of the insoluble drug material remains

Rates and orders of reactions...

- Reaction taking place in pharmaceutical suspensions are generally of pseudo zero-order type
- As the drug in solution decomposes more of it from the suspended particles goes into solution so that the concentration in the solution remains constant
eg. Aspirine suspension

Rates and orders of reactions...

Pseudo First order rxn

- In second-order rxn, if the concentration of one reactant is in such large excess that it virtually remains constant, then the rate of change of concentration follows first-order.
- Examples of pseudo first order rxn
 - Hydrolysis reactions (acid or base catalysed)
hydrolysis of ethyl acetate:
- $\text{CH}_3\text{COOEt} + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{COOH} + \text{EtOH}$

Rates and orders of reactions...

Methods for determining the order of reaction

☐ **Substitution method**

☐ **Graphical method**

☐ **Half life method**

Rates and orders of reactions...

Substitution method

- The data obtained from a kinetic experiment is substituted in the relevant integrated equation.
- The equation that yields a fairly constant value of K within the limits experimental variation indicates the order of the reaction

Rates and orders of reactions...

Graphical method

- The data obtained from a kinetic experiment is plotted in the relevant form to determine the order of a particular reaction
- **Zero-order** if straight line is obtained by plotting the concentration a against time t
- **First-order** if a plot of $\log c$ versus t yields a straight line, the reaction is first order.
- **Second-order** if a plot of $x/[a(a-x)]$ versus t yields a straight line provided the initial concentrations are equal

Rates and orders of reactions...

Half life method

- This involves the selection of a set of convenient initial concentrations and then determining the times taken to fall to half these values.
- If the reaction is zero order, then $C_0/2t_{1/2}$ would be a constant value (K).

For first order $0.693/t_{1/2}$ would be a constant k

- Similarly, for a second-order reaction $k = 1/C_0t_{1/2}$ would be constant.

Rates and orders of reactions...

Half life method....

Example. The Co and half life for the decomposition of a compound is tabulated below, Determine the order of the reaction?

Co	10	8	6	4	2
Co /2	5	4	3	2	1
t_{1/2} (h)	12	13	13	14	13

Rates and orders of reactions...

Half life method.....

Co	10	8	6	4	2
<i>K, zero Co/2t_{1/2}</i>	<i>0.42</i>	<i>0.31</i>	<i>0.23</i>	<i>0.14</i>	<i>0.08</i>
<i>K, 1st 0.693/t_{1/2}</i>	<i>0.057</i>	<i>0.053</i>	<i>0.053</i>	<i>0.0495</i>	<i>0.053</i>
<i>K, 2nd 1/Cot_{1/2}</i>	<i>0.008</i>	<i>0.009</i>	<i>0.013</i>	<i>0.018</i>	<i>0.038</i>

So the reaction is **first order** with a constant **k= 0.053/h**

Rates and orders of reactions...

Q. 1. The kinetics of decomposition of a drug in aqueous solution were studied using a series of solutions of different initial drug concentrations, C_0 . For each solution the time taken for half the drug to decompose (that is, $t_{0.5}$) was determined with the following results:

C_0 (mol dm ⁻³)	4.625	1.698	0.724	0.288
$t_{0.5}$ (min)	87.17	240.1	563.0	1414.4

Determine the order of reaction.

Degradation of drugs

- Drug substances and excipients have diverse molecular structures
 - susceptible to many degradation pathways.
- **chemical degradation pathways**
 - hydrolysis, dehydration, isomerization, oxidation, decarboxylation and elimination and photodegradation
- **physical change**
 - Vaporization, Crystallization of Amorphous Drugs, Transitions in Crystalline States, and Moisture Adsorption

Chemical degradation

Chemical degradation

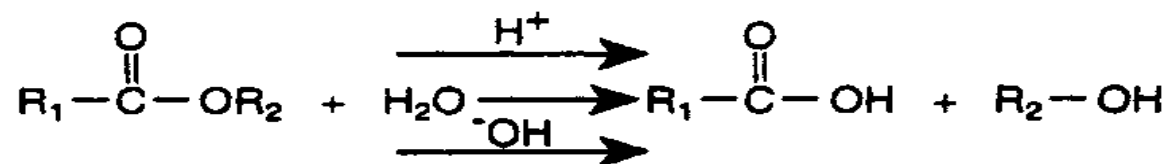
Hydrolysis (solvolysis)

- interaction of drug molecules with water molecule to yield breakdown product
- Most important
 - in system containing water such as emulsion, suspensions, solutions
 - for drug which are affected by trace of moisture in the form of water vapour from the atmosphere

Chemical degradation...

□ Esters hydrolysis

- Involves acyl-oxygen cleavage
- procaine, atropine, aspirin, tetracycline and physostigmine



□ Amides hydrolysis

- Involves the cleavage of the amide linkage to give an amine
- chloramphenicol, penicillin, cephalosporins, niacinamide, and barbiturates



Chemical degradation...

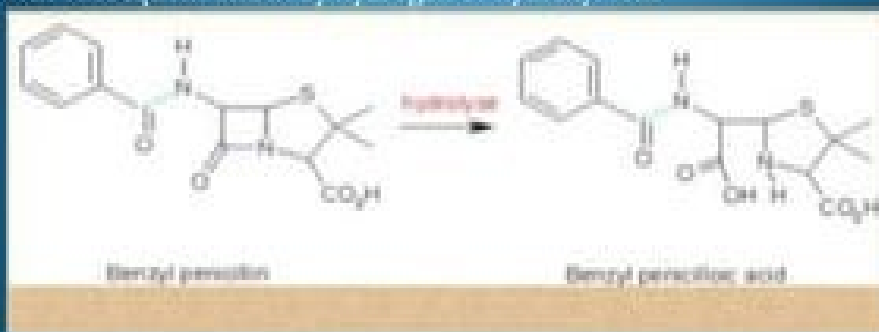
□ Ring hydrolysis

- Proceed by ring cleavage with subsequent attack by hydrogen or hydroxyl ions
- penicillins, cephalosporins, nitrazepam and chlordiazepoxide

Ring hydrolysis

RING hydrolysis:

Hydrolytic reactions in certain drugs proceed by ring cleavage with subsequent attack by hydrogen or hydroxyl ions



FOR EXAMPLE:

Cephalosporin , Penicillin , Nitrazepam .

Chemical degradation...

Protection against hydrolysis

- Solid dosage form may be prevented by
 - avoiding their contact with moisture at the time of manufacture,
 - packaging in suitable moisture resistant packs
 - storage in controlled humidity and temperature
 - incorporating a suitable desiccant in the pack such as silica gel bags

Chemical degradation...

- Liquid dosage form can be prevented by
 - Selecting an optimum pH for maximum stability and the formulation should be stabilized at this pH by inclusion of proper buffering agent
 - By altering the dielectric constant of the system by partial or complete replacement of water with non-aqueous solvents such as alcohol, glycerin and propylene glycol.

Chemical degradation...

- Addition of specific complexing agent like caffeine to the drug solutions
- By preparing insoluble derivatives of the drug which can then be formulated in the form of suspension.
- Formulating susceptible drugs such as penicillin and its derivative in the form of dry powder for reconstitution

Chemical degradation...

Oxidation

- In pharmaceutical dosage forms, oxidation is usually mediated through reaction with atmospheric oxygen under ambient conditions
- **Autoxidation** is the most common of oxidative degradation that occur in many of the pharmaceutical preparation
 - Only small amount of oxygen is required for initiating the r_{xn}

Chemical degradation...

- Many oxidation reactions are catalyzed by acids and bases
- Heavy metals like copper, iron, cobalt and nickel catalyse the oxidation rxn
- Heat and light further influence the kinetics of oxidative degradation process

Chemical degradation...

Protection against oxidation

- Use of Antioxidants

- antioxidants

- ✦ Block chain reaction by reacting with free radicals
- ✦ Inhibit autoxidations but **not effective in redox reactions**.
- ✦ Eg. Tocopherol, ascorbyl palmitate, propyl gallate

- Reducing agents

- ✦ are more readily oxidized than the drug or adjuvant,
- ✦ are effective in the redox rxn.
- ✦ Eg sodium metabisulphite, sodiumthiosulphate

Chemical degradation...

- Replacing oxygen in pharmaceutical containers with nitrogen or carbon dioxide.
- Storage at reduced temperatures
- Oxidation of fats and oils may be retarded by hydrogenation
- If oxidation is catalyzed by hydrogen or hydroxyl ion , the pH of optimum stability must be ensured

Chemical degradation...

Photodegradation

- sunlight or room light may cause substantial degradation of drug molecules.
- The energy from light radiation must be absorbed by the molecules to cause a photolytic reaction.
 - If this energy is sufficient to achieve activation, degradation of the molecule is possible

Chemical degradation...

- Photolysis reactions often are associated with oxidation
 - because the oxidation reactions frequently can be initiated by light.
 - But photolysis reactions are not restricted to oxidation
- Eg. phenothiazines, hydrocortisone, prednisolone, riboflavin, nifedipine, colchicine and chlorpromazine

Chemical degradation...

Protection against Photolysis

- use of coloured glass containers (amber glass excludes light of wavelength < 470 nm)
- storage in the dark place
- Packaging in cartons (physical barrier to light)
- opaque shells for capsules
- Coating tablets with a polymer film containing ultraviolet absorbers

Chemical degradation...

Dehydration

The elimination of a water molecule from structure of the cpd.

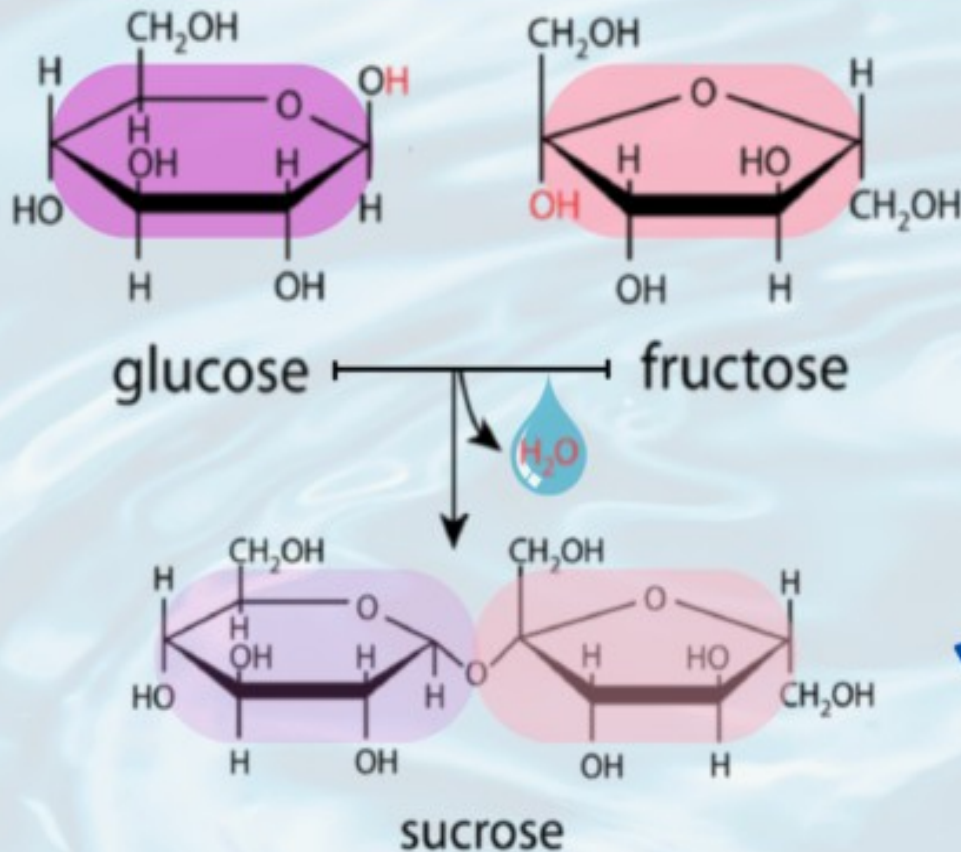
- In covalent (chemical) dehydration process, there is the formation of a double bond that can then participate in electronic resonance with neighboring functional groups

Eg. prostaglandin E2 and tetracycline

Sugars (glucose and lactose) are known to undergo dehydration to form 5-(hydroxymethyl) furfural.

Dehydration....

Dehydration reaction



Dehydration reaction

a form of biochemical reaction wherein water molecule forms as a byproduct.

THE DEHYDRATION SYNTHESIS OF SUCROSE -- WHERE GLUCOSE AND FRUCTOSE COMBINE AND WATER MOLECULE IS RELEASED

Chemical degradation...

- In physical dehydration processes water removal does not create new bonds but often changes the crystalline structure of the drug.
- Anhydrous compounds may have different dissolution rates compared to their hydrates
 - dehydration reactions involving water of crystallization may potentially affect the absorption rate of the dosage form.

Eg theophylline hydrate and ampicillin trihydrate

chemical degradation.....

Isomerization

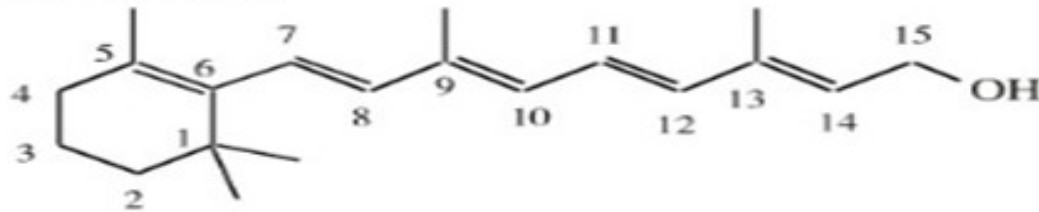
- Isomerization is the process of conversion of a drug into its **optical** or **geometric isomers**.
- Since the various isomers of a drug have different **activities**, such conversion may be regarded as a form of degradation, often resulting in a **serious loss of therapeutic activity**.

chemical degradation.....

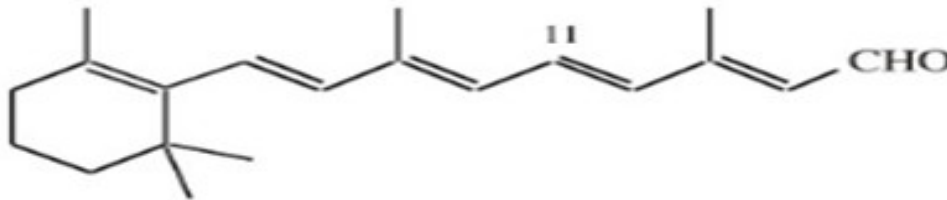
- Cis-trans isomerization may be a cause of loss of potency of a drug if the two geometric isomers have different therapeutic activities.
- For example, vitamin A (all-trans-retinol) is enzymatically oxidized to the aldehyde and then isomerized to yield 11-cis-retinal, which has a decreased activity compared with the all-trans molecule.

Chemical degradation...

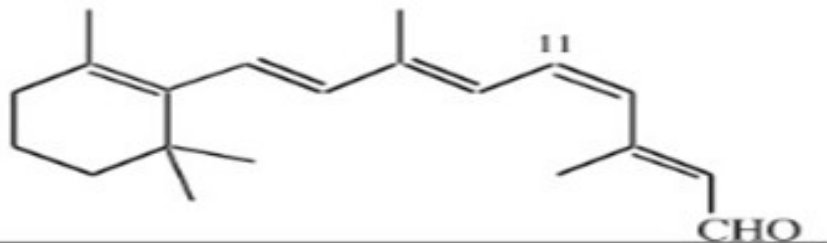
Retinol (vitamin A)



All *trans*-Retinal



11-*cis*-Retinal



Chemical degradation...

Elimination

- Drug substances having a carboxylic acid group are sometimes susceptible to decarboxylation
 - E.g. 4-Aminosalicylic acid under strongly acidic conditions
- Other elimination reactions have been reported for various drug substances
 - E.g. levothyroxine eliminates iodine

Chemical degradation...

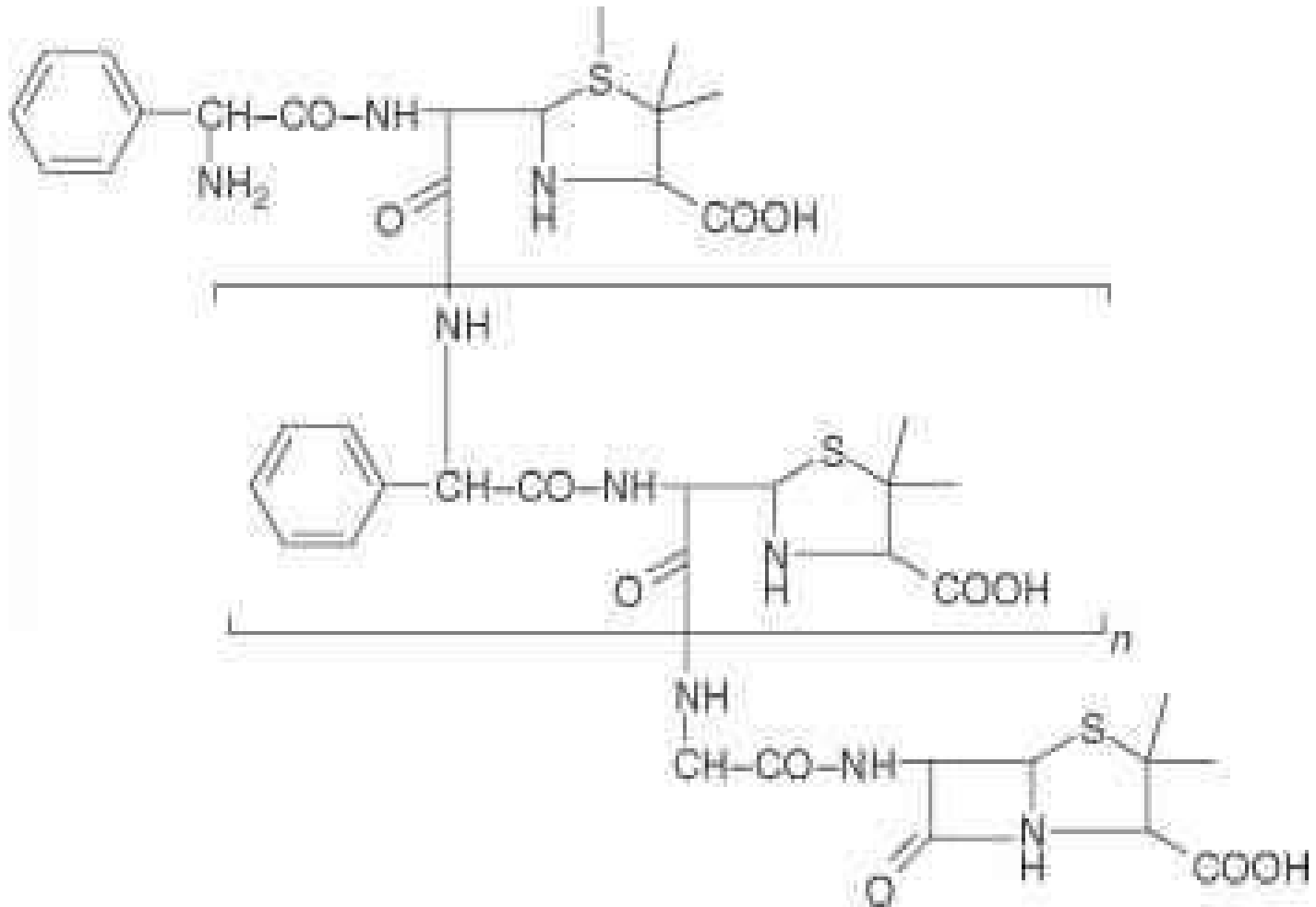
Polymerisation

- is the process by which two or more identical drug molecules combine together to form a complex molecule.
- a polymerisation process occurs during the storage of concentrated aqueous solutions of aminopenicillins, such as ampicillin sodium.
 - The reactive β -lactam bond of the ampicillin molecule is opened by reaction with the side-chain of a second ampicillin molecule and a dimer is formed. The process can continue to form higher polymers.
 - elicit penicilloyl-specific allergic reactions

Chemical degradation

Polymerization of ampicillin

p



Physical degradation

Physical Degradative Routes

Vaporization

- Some drugs and excipients possess sufficiently high vapor pressures at room temperature that their volatilization constitutes a major route of drug loss.
- Flavors, solvents and cosolvents (LMW alcohols) may be lost from the formulation
 - Volatility of solvents could leads to
 - potentially hazardous concentration of drugs.
 - Crystallization of drug and excipients.

Physical degradation...

- Eg. Nitroglycerin
- Reduction of vapor pressure can be achieved through dispersion of the volatile drug in macromolecules that can provide physicochemical interactions
 - The addition of macromolecules such as polyethylene glycol, polyvinylpyrrolidone, and microcrystalline cellulose allows for preparation of "stabilized" nitroglycerin sublingual tablets
 - A β -cyclodextrin-nitroglycerin tablet is currently being marketed

Physical degradation...

Adsorption (Sorption)

- Drug-plastic interaction is a major potential problem when intravenous solutions are stored in bags or infused via administration sets that are made from polyvinyl chloride (PVC)
 - up to 50% drug loss can occur after nitroglycerin is stored in PVC infusion bags for 7 days at room temperature.
 - more than 40% of a dose of quinidine gluconate was lost when the drug was administered with a conventional PVC intravenous administration set
- A diverse array of drugs, including diazepam, insulin, isosorbide dinitrate, and others, has shown substantial adsorption to PVC.

Physical degradation...

Crystallization of Amorphous Drugs

- Poorly water-soluble drugs formulated in their amorphous state to improve the solubility
 - because of the lower free energy of the crystalline state, amorphous substances tend to change to their more thermodynamically stable crystalline state with time.
- **Therefore, crystallization of amorphous drug substances may occur during long-term storage and may lead to drastic changes in the release characteristics of the drug and, hence, the absorption**

Physical degradation...

Transitions in Crystalline States

- Polymorphs are different crystalline forms of the same drug.
- have different free energy
 - transitions between polymorphs
- may alter critical properties of drugs because the solubility and dissolution rate of drug substances generally vary with changes in their crystalline form
- Temperature and humidity affect polymorphic transitions
- Examples:- Nitrofurantoin, Theophylline

Physical degradation.....

☐ Manifestations of Physical degradation/instability

▪ **Solutions:**

- ✓ Loss of original clarity
- ✓ Lose of original color
- ✓ Lose of original odor
- ✓ Lose of original test
- ✓ Change in viscosity
- ✓ Gas evolution
- ✓ Microbial growth
- ✓ Precipitation, etc

Physical degradation.....

■ Suspensions:

- ✓ Caking
- ✓ Crystal growth
- ✓ Lose of original color
- ✓ Lose of original odor
- ✓ Lose of original test
- ✓ Change in viscosity
- ✓ Microbial growth
- ✓ Change in drug dissolution
- ✓ etc,

Physical degradation.....

■ Emulsions:

- ✓ Creaming
- ✓ Sedimentation
- ✓ Flocculation
- ✓ Coalescence
- ✓ Breaking
- ✓ Phase inversion

- ✓ Lose of original color
- ✓ Lose of original odor
- ✓ Lose of original test
- ✓ Change in viscosity
- ✓ Gas evolution
- ✓ Microbial growth
- ✓ etc,

Factors affecting stability of drugs

Environmental factors

■ Temperature

- The speed of many reactions increases about two to three times with each 10 °C rise in temperature.
- For example, there is pronounced increase in hydrolysis rate of many drugs by increase in temperature.

Factors affecting stability of drugs.....

Temperature.....

- according to classic collision theory reaction rates are proportional to the number of collisions per unit time.
- Because the number of collisions increases as the temperature increases, the reaction rate is expected to increase with increasing temperature.
- The speed of many reactions increases about 2 to 3 x with each 10 degree rise in T.

Factors affecting stability of drugs ...

Temperature...

- The effect of temperature on reaction rate is given by the Arrhenius equation

$$k = Ae^{-E_a/RT} \quad (26)$$

or

$$\log k = \log A - \frac{E_a}{2.303 RT} \quad (27)$$

Factors affecting stability of drugs ...

Where

- K is the specific reaction rate constant
 - A is a constant known as the Arrhenius factor
 - E_a is the energy of activation
 - R is the gas constant , 1.987 calories/deg mole or 8.314J/K mole
 - T is the absolute temperature
-
- E_a and A can be obtained by determining K at several temperatures and plotting $1/T$ against $\log K$
 - The slope of the line obtained is $-E_a/2.303R$ and the intercept on the vertical axis is $\log A$

Factors affecting stability of drugs ...

- E_a can also be obtained by writing eq. 27 for a temperature T_1 as

$$\log k_1 = \log A - \frac{E_a}{2.303R} \frac{1}{T_1} \quad (28)$$

- And for another temperature T_2 as

$$\log k_2 = \log A - \frac{E_a}{2.303R} \frac{1}{T_2} \quad (29)$$

- Subtracting the eq. 28 from the eq. 29 yields

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left(\frac{T_2 - T_1}{T_2 T_1} \right) \quad (30)$$

Factors affecting stability of drugs ...

Q.1. The rate constant k_1 for the decomposition of 5-hydroxymethylfurfural at 120°C (393 K) is 1.173 hr^{-1} or $3.258 \times 10^{-4} \text{ sec}^{-1}$ and k_2 at 140°C (413 K) is 4.860 hr^{-1} . What is the activation energy, E_a , in kcal/mole and the frequency factor, A , in sec^{-1} for the breakdown of 5-HMF within this temperature range?

Q.2. The first-order rate constant for the hydrolysis of sulfacetamide at 120°C is $9 \times 10^{-6} \text{ S}^{-1}$ and the activation energy is 94 kJ mol^{-1} . Calculate the rate constant at 25°C

Factors affecting stability of drugs ...

Light

- Ultraviolet light is the most usual cause of photodegradation.
- Photochemical rxn do not depend on temperature for activation of molecules and hence the rate of activation for such rxn is independent of T
- Are complex rxns and proceed by a series of steps

Factors affecting stability of drugs...

Light

- Photolabile drugs are usually stored in containers which exclude ultraviolet light.
- Amber glass is particularly effective in this respect because it excludes light of wave length of less than about 470 nm.
- As added precaution, it is always advisable to store photolabile drugs in dark place.

Factors affecting stability of drugs...

■ Moisture

- Moisture, which is water existing in the air when adsorbed on to many drug products cause hydrolysis of drugs.

■ Oxygen

- Oxygen is involved in many oxidative schemes.
- Formulations that are shown to be susceptible to oxidation can be stabilized by replacing the oxygen in the storage container with **nitrogen** or **carbondioxide**, or by adding **antioxidants**.

Factors affecting stability of drugs...

□ Formulation factors

pH

- pH is perhaps the most important parameter which affects hydrolysis rate of drugs in liquid formulations.
- Solutions of a number of drugs undergo accelerated decomposition on the addition of acids or bases
- It also affects oxidative degradation of some drugs in solution. For example, oxidation of prednisolone

Factors affecting stability of drugs ...

- If the drug solution is buffered, the decomposition may not be accompanied by appreciable change in the concentration of acid or base so that the reaction can be considered to be catalyzed by hydrogen or hydroxyl ions.
- When the rate law for such an accelerated decomposition is found to contain a term involving the concentration of hydrogen ion or the concentration of hydroxyl ion, the rxn is said to be subject to specific acid-base catalysis.

Factors affecting stability of drugs ...

Solvent

- The effect of solvents on the rate of decomposition of drugs is related to the relative solubility of reactants and the products in the given solvents
 - Polar solvents tend to accelerate rxns in which the products formed are more polar than the reactants
 - If the products formed are less polar than the reactants then the rxn proceeds better in the solvents of relatively low polarity

Factors affecting stability of drugs ...

Ionic strength

It is a measure of the concentration of ions in that solutions

- The effect of ionic strength of a solution on the rate of degradation may be expressed in the form of the following equation:

$$\log k = \log k_0 + 1.02 z_A z_B \sqrt{\mu}$$

(31)

where

- z_A and z_B are the charge numbers of the two interacting ions
- K is the degradation rate constant for the reaction
- K_0 is the rxn rate constant at infinite dilution
- μ is the ionic strength of the solution

Factors affecting stability of drugs ...

- According to eq. 31 , increase in the ionic strength of the solution
 - decrease the rate of rxn b/n oppositely charged ions and
 - Increase the rate of rxn b/n similarly charged ions
- Rxn b/n ions and neutral molecules are generally not affected by to significant extent by change in ionic concentration

Factors affecting stability of drugs...

- **Excipients**
- **Aspirin** and **hydrocortisone** decompose in several PEGs which are often used as suppository bases.
- Some excipients absorb moisture.
 - **Starch** and **povidone** absorb water when stored at high relative humidity. As the result they affect stability of products which contains them.

Factors affecting stability of drugs....

- Other factors which affect stability of drugs:
 - manufacturing process,
 - nature of the container-closure system
 -

Stability study

- The purpose of stability testing is to provide evidence on how the quality of a **drug substance (active ingredient) or drug product (formulation) varies with time under the influence of a variety of environmental factors**
 - **temperature, humidity, and light.**
- Stability study is done for new formulations before being marketed.

Stability study

- Permits the establishment of recommended **storage conditions, retest periods, and shelf lives**
- Stability studies should include **testing of those attributes of the drug substance or drug product that are susceptible to change during storage and are likely to influence quality, safety, and efficacy**

Stability study...

- The testing should cover, as appropriate, **the physical, chemical, and microbiological attributes, preservative content and functionality test**
- Tests on chemical , physical and microbiological stability of drug substances and excipients involve application of **a suitable stress or challenge** such as temperature, light, RH, and microorganism, over a period of time and assess its effects.

Stability study...

Climatic zone

- According to ICH (international conference on harmonization) and WHO guideline on stability studies, the world has been divided in to four zones taking in to account the prevalent annual climatic conditions of temperature and humidity
 - Zone I- temperate
 - Zone II – subtropical with possible high humidity
 - Zone III- hot, dry
 - Zone IV –hot, humid

Stability study...

Stability testing protocols

- Stability testing requires the careful design of protocols which must define clearly the following:
 - The temperature and humidity for storage
 - Storage time before sampling
 - The number of batches to be sampled
 - The number of replicates within each batch
 - A suitable light challenge
 - Details of assay
 - container

Some parameters to be studied during stability study for dosage forms

Dosage form	Some parameters to be studied
Oral solution	Assay, formation of precipitate, clarity of solutions, pH, viscosity, extractables, level of microbial contamination
Suspension	Assay, caking, crystal growth, pH, viscosity, extractables, level of microbial contamination, redispersibility, rheological properties and mean size and size distribution of particles
Emulsions	Assay, phase separation, pH, viscosity, level of microbial contamination, and mean size and size distribution of dispersed globules
Ointments, creams, paste, gel,	Assay, clarity, homogeneity, pH, consistency, viscosity, particle size distribution (for insoluble drugs), level of microbial contamination

Stability studies...

Type of stability study

- Major types of stability testes are

- Long term stability test

- Accelerated stability test

- Stress stability test

Stability study...

Long term stability (Real time studies)

- ICH guideline Q1A(R2) define long term studies as stability studies under the recommended storage conditions for the retest period or shelf life proposed or approved for labeling
- WHO guideline defines long term testing as the stability testing during and beyond expected shelf life under storage conditions in the intended market.

Stability study...

- **Long term stability (Real time studies).....**
- Conducted in the laboratory under controlled stresses similar to those likely to be encountered during storage.
- Minimum of 12 months data
- For product stored at room temperature : $25 \pm 2^{\circ}\text{C}$ and RH of $60 \pm 5\%$ (zone I and II) and $30 \pm 2^{\circ}\text{C}$ and RH $65 \pm 5\%$ (zone III and IV)
- For drug substances and drug products recommended to be stored in a refrigerator: $5 \pm 3^{\circ}\text{C}$

Stability study...

Long term stability (Real time studies).....

For drug substances and drug product to be stored in a freezer: $-20 \pm 5^{\circ}\text{C}$

Testing frequency

- For long term storage condition should be **every 3 months over the first year**, **every 6 month over the second year** and **annually thereafter through the proposed shelf life**

Stability study...

Accelerated stability study

- Use of exaggerated conditions of temperature, humidity, light & others.
- Accelerated stability tests are done:
 - To detect the deterioration of the product rapidly, in order to select the best formulation;
 - To predict shelf life;
 - To provide rapid means of quality control.

Stability study...

Accelerated storage condition

- For drug substance and drug products stored at room temperature: 40 ± 2 °C with 75 ± 5 % RH (for all zone)
- For drug substance and drug products intended to be stored in a refrigerator : 25 ± 2 °C and RH of 60 ± 5 %
- Accelerated testing can help to identify **the likely degradation products**, which can in turn help establish the **degradation pathways** and the **intrinsic stability** of the molecule.

Stability study...

Accelerated storage condition

Testing frequency

- At the accelerated storage condition, a minimum of three points, including the initial and final time points (0, 3 and 6 months)
- When expectation exists that results from accelerated testing are likely to approach significant change criteria, increased testing should be conducted by including a fourth time point in the study design.

Stability study.....

What are the 'significant changes'?

- Significant changes at accelerated condition is defined as
 - ✓ A 5% or more loss in potency from the initial assay value
 - ✓ Any specific degradant exceeding its specification limit
 - ✓ The product exceeding its pH limits
 - ✓ Failure to meet specifications for appearance, color, phase separation, resuspendability, caking, hardness, etc.

Stability study...

Stress stability study

- Stress testing is used to elucidate the intrinsic stability of the *drug substance*
- *is part of the* development strategy and is normally carried out under more severe conditions than those used for accelerated testing.
- The testing typically includes the effects of temperature[in 10°C increments (e.g., 50°C–60°C) above that for accelerated testing], humidity (e.g., 75% relative humidity or greater) where appropriate, oxidation, and photolysis on the drug substance.

Prediction of shelf life...

Shelf life

- The time period during which the dosage form is supposed to retain its original qualities
- Steps involved in prediction of shelf life
 - The preparation is divided into d/f portions and each portion is stored at different elevated T such as 40, 50, 60 and 70 °C
 - Samples from each portion are withdrawn at various intervals of time and the remaining concentration of the active ingredients is measured

Prediction of shelf life...

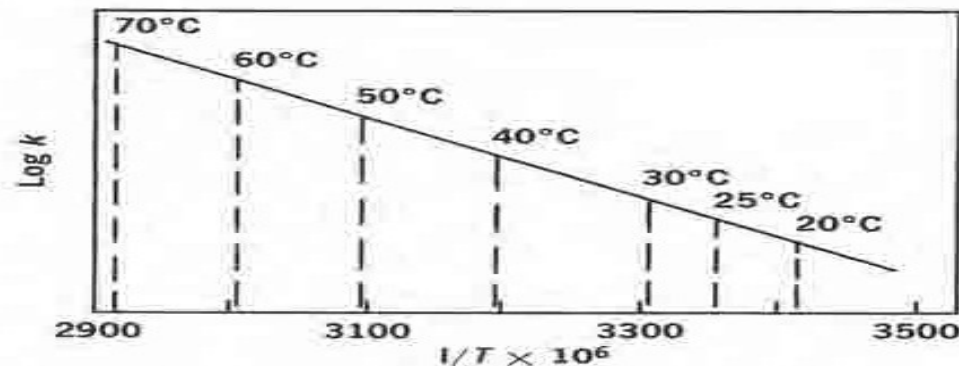
- The order of rxn is determined by a suitable methods such as graphic, half-life or substitution method
- From the slopes of the lines, the reaction rate constant K for the degradation at each of the elevated temperatures is calculated.
- K for the degradation at room temperature is determined by applying Arrhenius r/ship. It may also be obtained from the plot of $\log k$ values against the reciprocal of absolute T and extrapolating the curve to 25 °c

Prediction of shelf life...

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left(\frac{T_2 - T_1}{T_2 T_1} \right)$$

The k value is then substituted in the appropriate rate equation and an estimate is obtained for shelf life of the product

- Calculation are then carried out to determine the amount of drug to be added in excess to compensate for the loss in concentration



Prediction of shelf life...

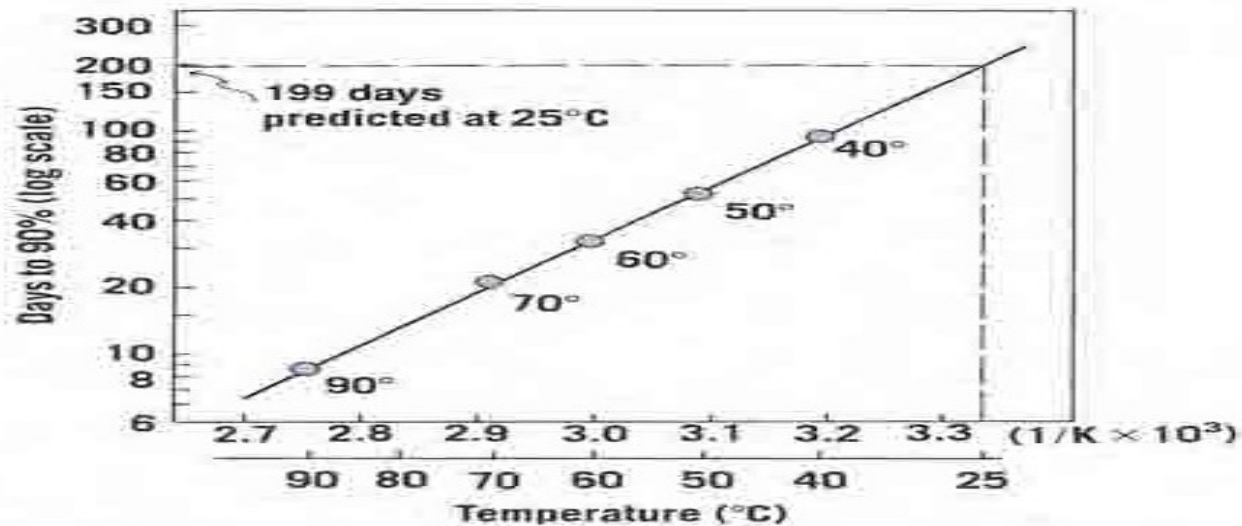
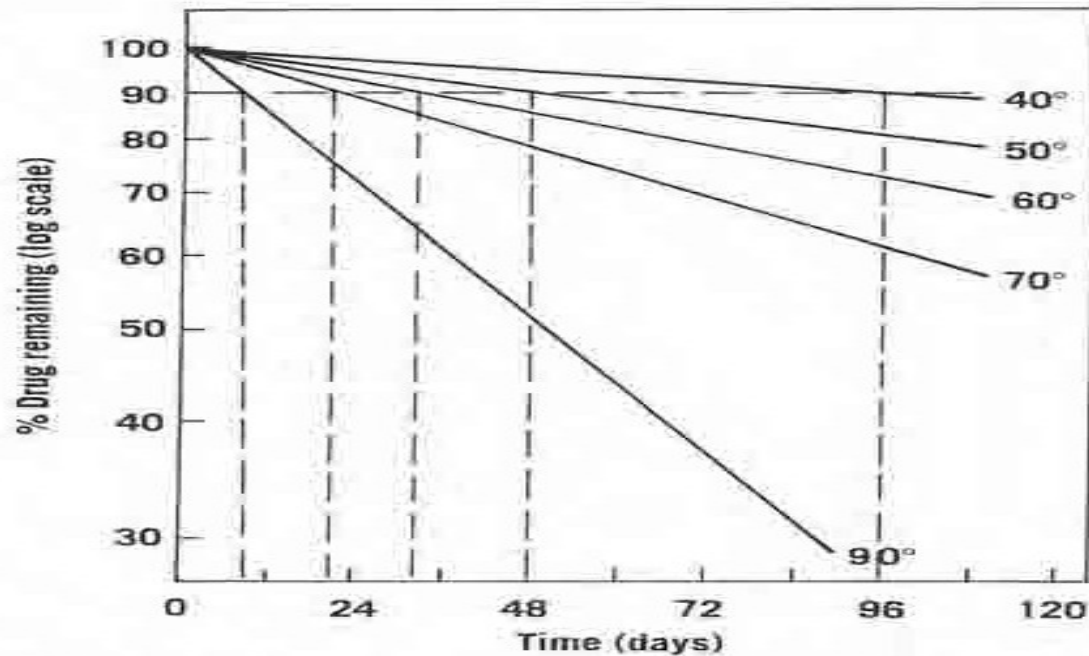
- A similar method with some modifications has also been suggested for the determination of shelf life
 - Log % of drug remaining is plotted against time (in days)
 - From the graph, the time required for the concentration to fall to 90% of the original value at different T is obtained

Prediction of shelf life...

- Then the log time to 90% is plotted against the reciprocal of T (either Celsius or Kelvin)
- The straight line obtained is extrapolated to lower T (25 °c) and from this the number of days required for the concentration to fall to 90% of its original value is read directly at 25 °c

Prediction of shelf life...

• p



Exercise 1

The initial concentration of a drug which decomposes according to 1st order kinetics, is 94 units/ml. the specific decomposition rate K obtained from an Arrhenius is 2.09×10^{-5} per sec at room T (25 °C). Previous experiments have shown that below 45 units/ml, the drug is not fit for use. What expiry date should assigned to the product? Ans 3.52×10^4 hr=4 year

Exercise 2

A solution of drug contained 500 mg/mL when prepared. It was analysed after 40 days and was found to contain 300 mg/mL. Assume the decomposition is zero order,

- ✓ Calculate the zero order rate constant
- ✓ At what time will the drug decompose to one-half of its original concentration?
- ✓ t_{90} (shelf-life)

Exercise 3

2. The initial concentrations of both ethyl acetate and sodium hydroxide in the mixture were 0.01000 M. The change in concentration, *of alkali during 20 min was 0.00056 mole/liter*

Compute

(a) the rate constant

(b) the half-life of the reaction